

CMPM 120 – Game 4(a) Final Game Design Document

Title

Moonlight Factory

Group Members

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Design Brief

Moonlight Factory is a 2D platformer game created using the Phaser game library. The project expands upon my previous CMPM 120 platformer assignment by adding additional levels, new gameplay mechanics, improved game feel, and more polished audiovisual feedback.

The player explores a dangerous industrial facility while avoiding hazards, navigating platforming challenges, and reaching the exit of each level. The game focuses on responsive movement, readable level design, and a small but polished gameplay experience.

The gameplay loop is simple: the player moves through platforming sections, avoids traps and hazards, interacts with moving elements in the environment, and progresses through multiple levels of increasing difficulty. The final version of the game will include multiple scenes such as a title screen, gameplay levels, and an ending/credits screen.

One of the main goals of this project is to improve the overall “game feel” of the platformer experience. This includes refining jump movement, improving player feedback through sound effects and particles, and making level transitions feel smoother and more complete. I also want to improve my understanding of Phaser scene management and code organization by structuring the game into reusable components and organized files.

The visual style of the game will use primarily existing pixel-art asset packs and sound assets. Original art created specifically for this project will remain below the course limit of 10% of all assets used.

Overall, the project is intended to be a small but polished platformer experience that demonstrates improvement in Phaser programming, game design, and technical organization.

Learning Objectives

1. Reinforce my understanding of Phaser tilemaps, collisions, and platformer mechanics.
2. Learn how to organize a larger Phaser project using multiple scenes and reusable code structures.
3. Improve platformer “game feel” through movement tuning, particles, sound effects, and visual feedback.
4. Design multiple playable levels with increasing difficulty and clearer level progression.
5. Practice implementing additional gameplay mechanics such as moving platforms, hazards, or checkpoints.

Self-Evaluation Rubric

Rubric Description	Number of Points	How Points Are Assigned
Multiple playable levels	3	3 points if the game includes at least 3 playable levels with clear progression and transitions.
Improved platformer game feel	3	3 points if movement, jumping, hazards, and player feedback feel responsive and polished.
New gameplay mechanics	3	3 points if the game adds at least two meaningful gameplay features beyond the original platformer assignment.
Audio and visual polish	3	3 points if the game includes sound effects, background music, particles, animations, or transition effects.
Code organization and structure	3	3 points if the project uses organized scenes, reusable code, meaningful variable names, and clean file structure.